Smart City Bike Sharing Analysis

Data Transformation, Modelling & Visualisation with Power BI

Project detail	Description
Project title	Analyzing Smart City Bike Sharing Data using Power BI and Power Query
Tools used	Power BI Desktop, Power Query Editor, DAX
Domain	Smart city public bike sharing
Report date	November 2026

1. Project Overview and Objective

Overview

- This project analyses a smart city bike-sharing dataset that records the status and bike availability of public bike stations.
- The stations belong to 25 bike-sharing contracts, and each contract is the bike system of one city or area.
- The raw file was cleaned and transformed in Power Query and split into one fact table and four dimension tables.
- A star-schema data model was built in Power BI, DAX measures were written for the key metrics, and an interactive dashboard was designed to turn the raw data into actionable urban mobility insights.

Objective

- The primary objective of this project is to apply data analytics techniques using Power Query and Power BI, covering data cleaning, transformation, data modelling, DAX calculations and interactive visualisation.
- The analysis aims to identify station performance patterns, evaluate bike availability, monitor operational issues and highlight areas requiring improvement through descriptive, diagnostic, predictive and prescriptive analysis.

2. Data Sources

Item	Details
Dataset	Bike station sharing data (bike-stations-sharing-data CSV)
Source description	Public bike-sharing station records: station name, address and location, contract, station status, number of bike stands, available bikes, available stands and last update
Timeline	Last-update dates from January 2022 to November 2025
Size	2,856 stations across 25 contracts
Domain	Smart city public bike sharing (transportation analytics)

3. Problem Statement

Public bike-sharing systems generate continuous data from thousands of stations across different cities. This project analyses that station data to understand station performance, usage efficiency and operational patterns across contracts.

- To understand the size of the network: which contracts have the most stations and where the stations are located.
- To study station operations: how many stations are open or closed, which contracts account for most closures, and whether closures depend on network size.
- To evaluate bike availability across contracts and identify contracts with a shortage of bikes.
- To examine how the average bikes per station and the number of closed stations change over time.
- To recommend actions: where bikes should be moved and which contracts should repair stations first.

4. Attribute (Column / Features) Details

The raw data was split into one fact table and four dimension tables. The attributes of each table are listed below.

Fact table: Bike_availability_fact

Attribute name	Data type	Description
Station ID	Text	Foreign key linking to Station_dim
Contract ID	Text	Foreign key linking to Contract_dim
Status ID	Text	Foreign key linking to Status_dim
Date	Date	Date of the last update; links to Date_dim
Bike Stands	Whole number	Total number of bike stands at the station
Available Bikes	Whole number	Number of bikes available at the station
Available Bike Stands	Whole number	Number of empty stands available for returning a bike

Dimension table: Station_dim

Attribute name	Data type	Description
Station ID	Text	Unique identifier for each station (primary key), for example S001
Station name	Text	Name of the bike station
Address	Text	Street address of the station
Latitude	Decimal number	Latitude of the station; used for the map
Longitude	Decimal number	Longitude of the station; used for the map

Dimension tables: Contract_dim and Status_dim

Attribute name	Data type	Description
Contract ID	Text	Unique identifier for each contract (primary key), for example C001
Contract Name	Text	Name of the bike-sharing system; used as the location in the analysis
Status ID	Text	Unique identifier for each status (primary key), for example ST01
Status	Text	Station status: OPEN or CLOSED

Dimension table: Date_dim

Attribute name	Data type	Description
Date	Date	Unique date (primary key)
Year	Whole number	Calendar year
Month	Whole number	Month number (1 to 12)
Month name	Text	Name of the month, for example January
Year month	Text	Year and month label, for example 2025-Jan

5. Tools & Technologies

Tool	Purpose
Power Query Editor	Data import, profiling, cleaning, transformation and splitting the data into fact and dimension tables
Power BI Desktop	Data modelling, relationships, report design and the interactive dashboard
DAX	Measures for KPIs such as availability %, closed station % and averages
Power BI visuals	Cards, column and bar charts, funnel, map, donut, scatter, waterfall, line charts with trend lines, tables with conditional formatting and slicers

6. Data Pre-Processing (Power Query)

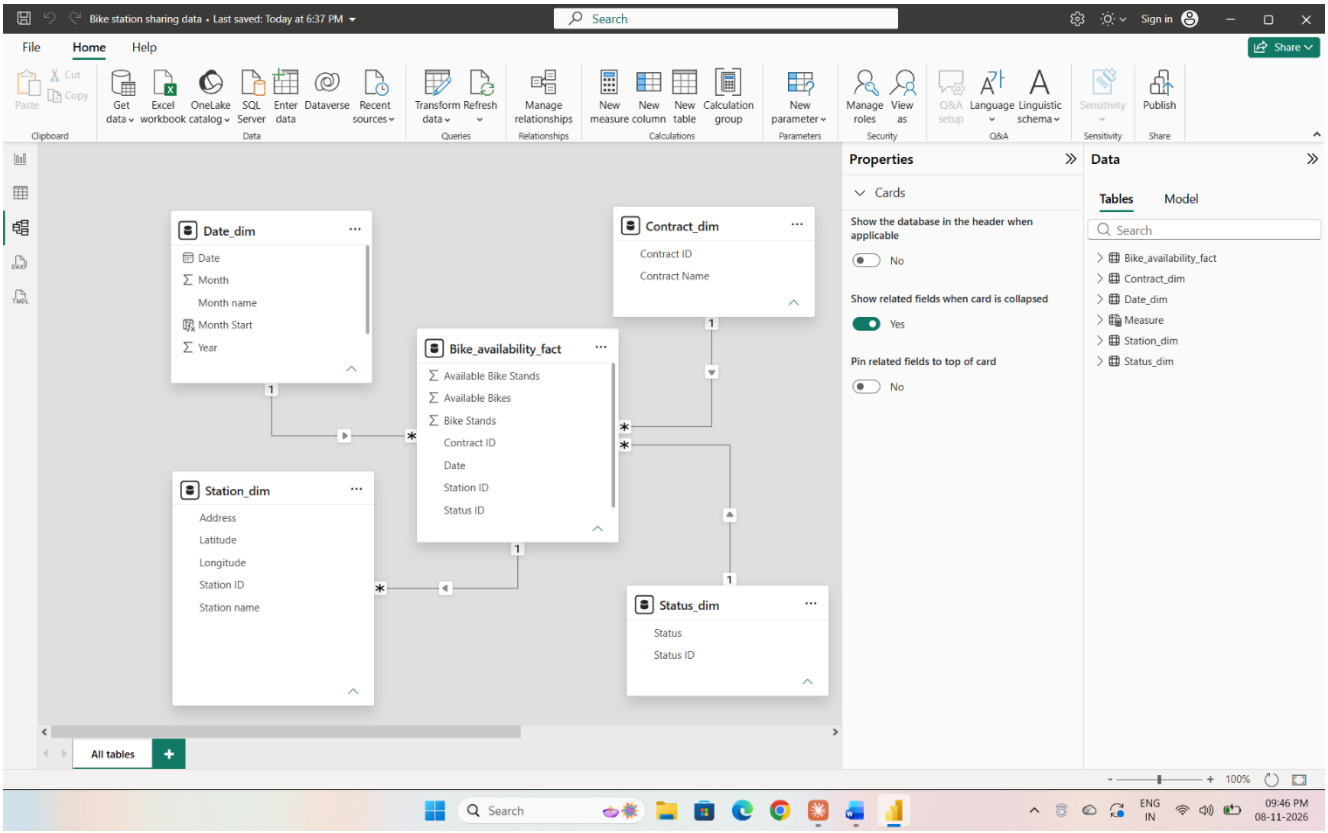
The CSV file was imported with Get Data and opened in Transform Data (Power Query Editor).
Each step is recorded in Applied Steps, which documents the transformation and keeps it repeatable.

Step	Action	Purpose	Category
1	Imported the CSV file (Get Data, Text/CSV)	Load the raw bike station data	Data import
2	Profiled the data with Column quality, Column distribution and Column profile	Check for empty values, errors and duplicates	Data cleaning
3	Set correct data types (text, whole number, decimal, date)	Make sure calculations and visuals work correctly	Data cleaning
4	Split the Last Update column into Date and Time	Enable date-based analysis and the Date_dim table	Transformation
5	Removed unnecessary columns	Keep only the columns needed for the analysis	Transformation
6	Created Station_dim, Contract_dim and Status_dim with unique IDs and removed duplicate rows	Build the dimension (lookup) tables	Fact and dimension
7	Created Bike_availability_fact with the ID keys, date and numeric columns	Build the central fact table	Fact and dimension
8	Created Date_dim with Date, Year, Month, Month name and Year month	Support monthly and yearly analysis	Fact and dimension
9	Sorted and filtered the data while checking the results	Focus on the relevant records	Filtering and sorting

7. Data Modelling and DAX (Power BI)

Data model

A star schema was built with Bike_availability_fact in the centre and four dimension tables around it. Every relationship is many-to-one from the fact table to each dimension table (many-to-one from the dimension side) with a single filter direction.



Fact table column	Dimension column	Cardinality	Filter direction
Bike_availability_fact[Station ID]	Station_dim[Station ID]	Many to one	Single
Bike_availability_fact[Contract ID]	Contract_dim[Contract ID]	Many to one	Single
Bike_availability_fact[Status ID]	Status_dim[Status ID]	Many to one	Single
Bike_availability_fact[Date]	Date_dim[Date]	Many to one	Single

DAX measures

Measure	DAX formula	Purpose	Result
Total Stations	`DISTINCTCOUNT(Bike_availability_fact[Station ID])`	Number of stations	2,856
Total Contracts	`DISTINCTCOUNT(Contract_dim[Contract Name])`	Number of contracts	25
Total Bike Stands	`SUM(Bike_availability_fact[Bike Stands])`	Total capacity	57.20K
Total Available Bikes	`SUM(Bike_availability_fact[Available Bikes])`	Bikes available	19.65K

Measure	DAX formula	Purpose	Result
Total Available Bike Stands	`SUM(Bike_availability_fact[Available Bike Stands])`	Empty stands	32.12K
Open Stations	`CALCULATE([Total Stations], Status_dim[Status] = "OPEN")`	Stations open	2.65K
Closed Stations	`CALCULATE([Total Stations], Status_dim[Status] = "CLOSED")`	Stations closed	206
Open Station %	`DIVIDE([Open Stations], [Total Stations])`	Share of open stations	92.79%
Closed Station %	`DIVIDE([Closed Stations], [Total Stations])`	Share of closed stations	7.21%
Bike Availability %	`DIVIDE([Total Available Bikes], [Total Bike Stands])`	Share of stands holding a bike	34.34%
Avg Bikes per Station	`AVERAGE(Bike_availability_fact[Available Bikes])`	Average bikes per station	6.88
Avg Stands per Station	`AVERAGE(Bike_availability_fact[Bike Stands])`	Average stands per station	20.03

8. Analysis and Visualisations (Power BI)

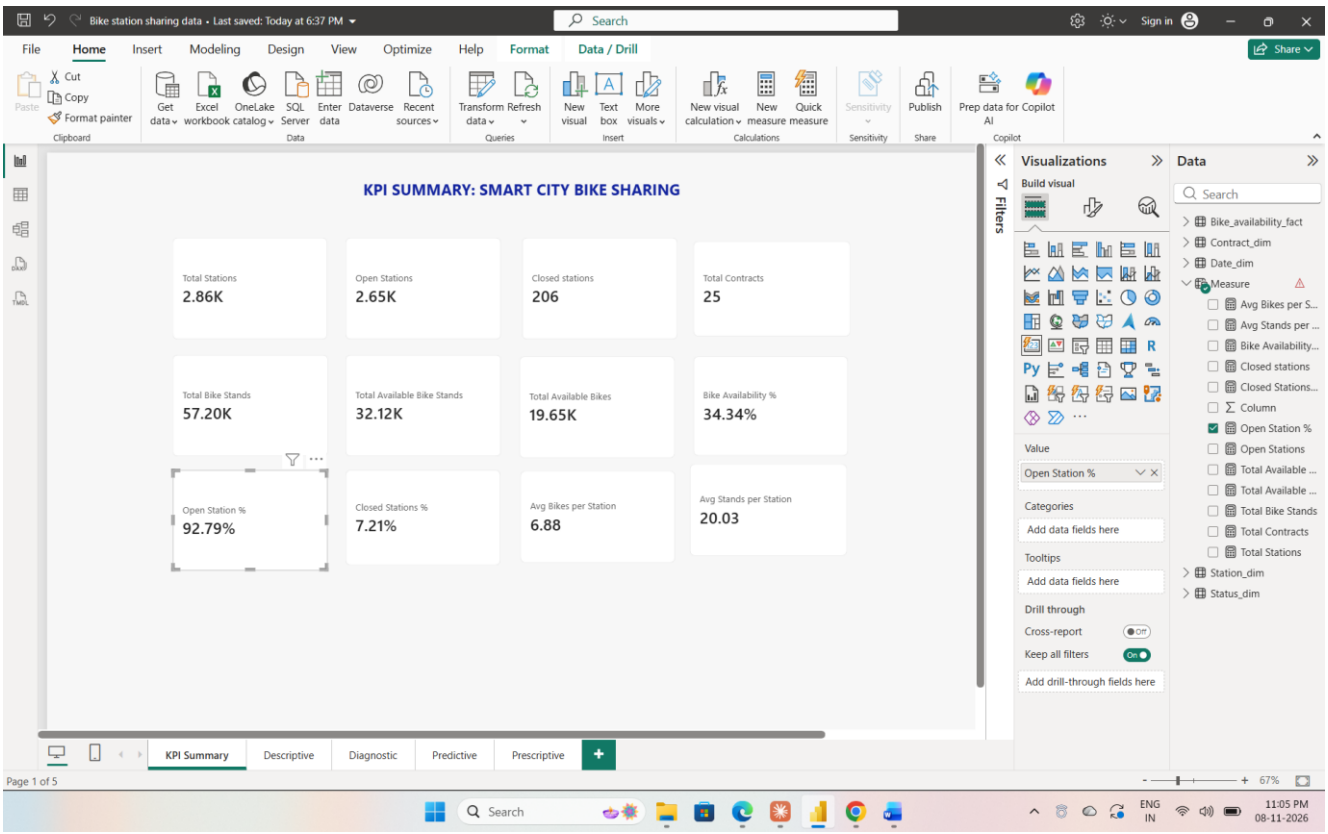
The report has five pages: a KPI summary and one page for each type of analysis.
Each visual has a clear title and data labels, and every page has the same four slicers.

Report page	Visuals (titles)
KPI Summary	12 KPI cards: Total Stations, Open Stations, Closed Stations, Total Bike Stands, Total Available Bike Stands, Total Available Bikes, Bike Availability %, Avg Bikes per Station, Avg Stands per Station, Closed Stations %, Open Station %, Total Contracts
Descriptive Analysis	5 KPI cards; Stations by Contract (column chart); Top 10 Contracts by Number of Stations (funnel chart); Bike Station Locations (map); Open vs Closed Stations (donut chart)
Diagnostic Analysis	Stations vs Closed Stations by Contract (scatter chart); Average Bikes per Station by Contract (bar chart); Closed Stations by Contract (waterfall chart)
Predictive Analysis	Average Bikes per Station by Month (line chart with trend line); Closed Stations Trend by Month (line chart with trend line)
Prescriptive Analysis	Contracts Needing More Bikes (table with conditional formatting); Contracts Needing Station Repairs (table with conditional formatting)

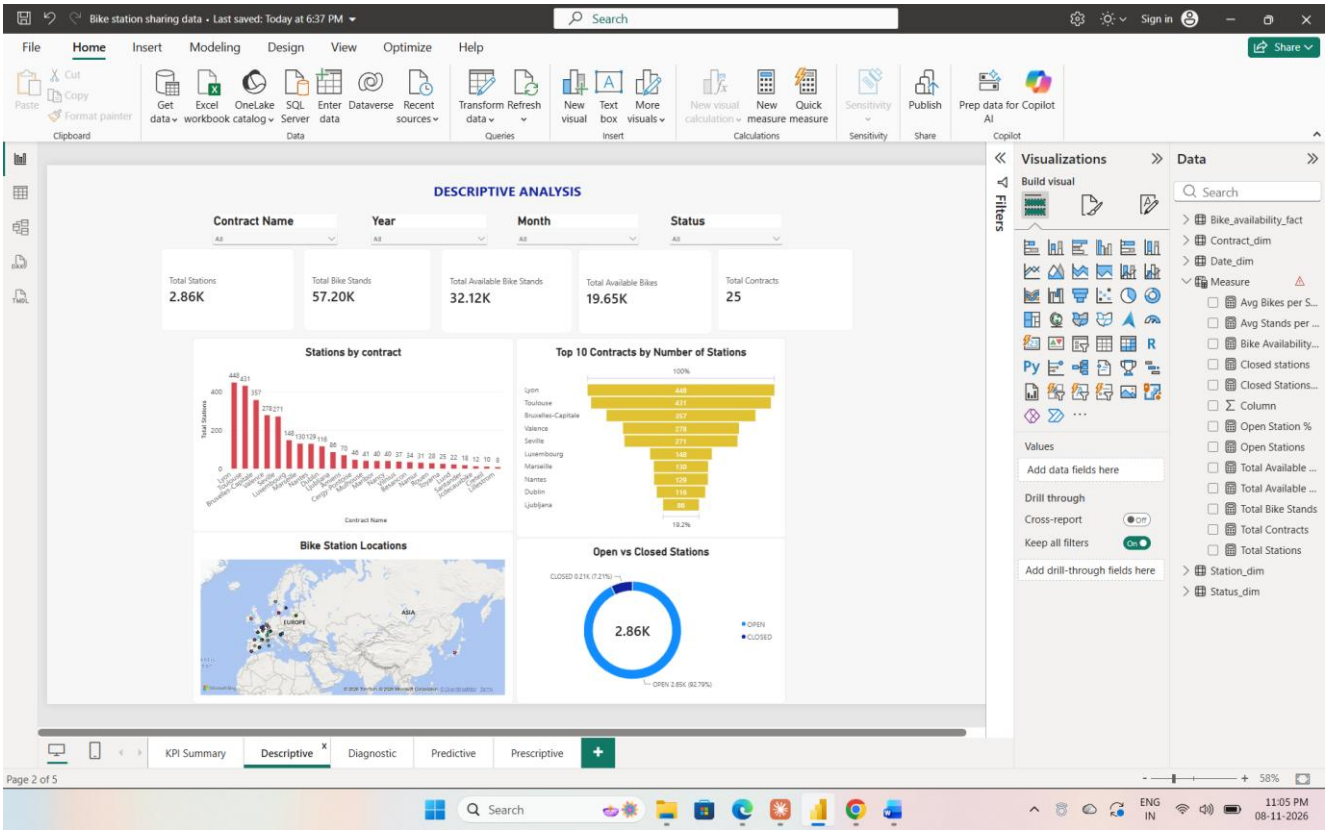
Interactive features

- Slicers for Contract Name, Year, Month and Status on every page, so the user can filter the whole page.
- A Top 10 filter on the funnel chart to show only the largest contracts.
- Conditional formatting (red, orange and green) in the prescriptive tables to highlight priority contracts.
- Data labels, axis titles and clear visual titles for easy reading.

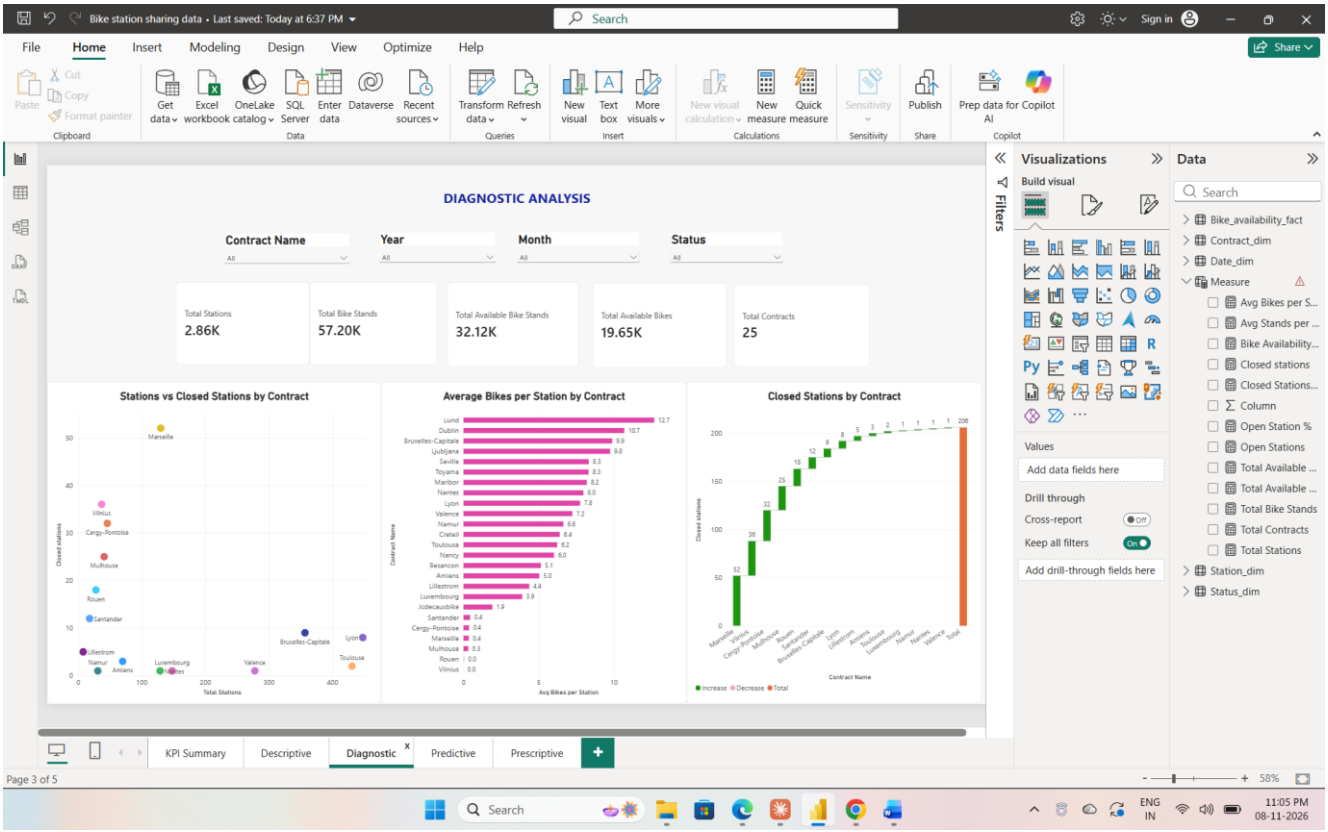
Dashboard Visualizations



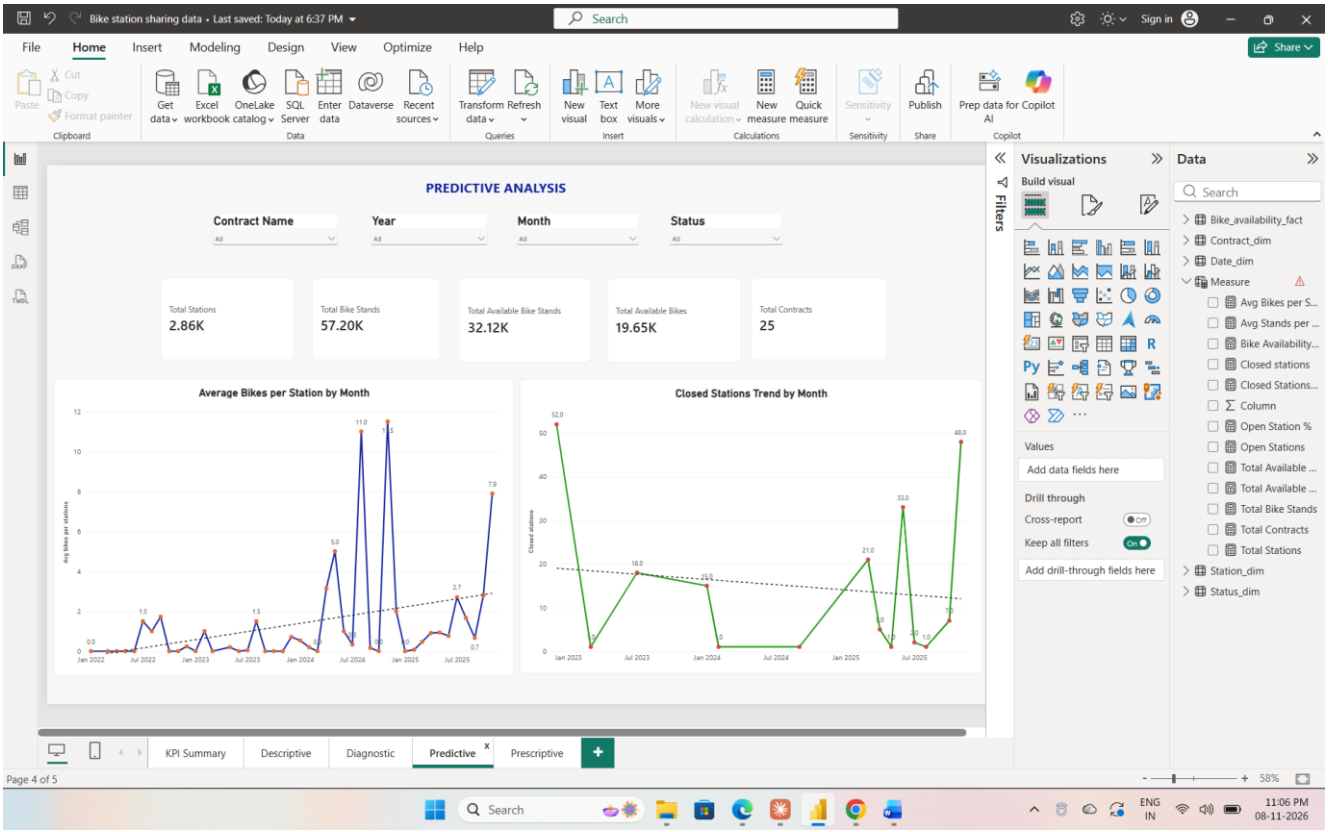
KPI Summary



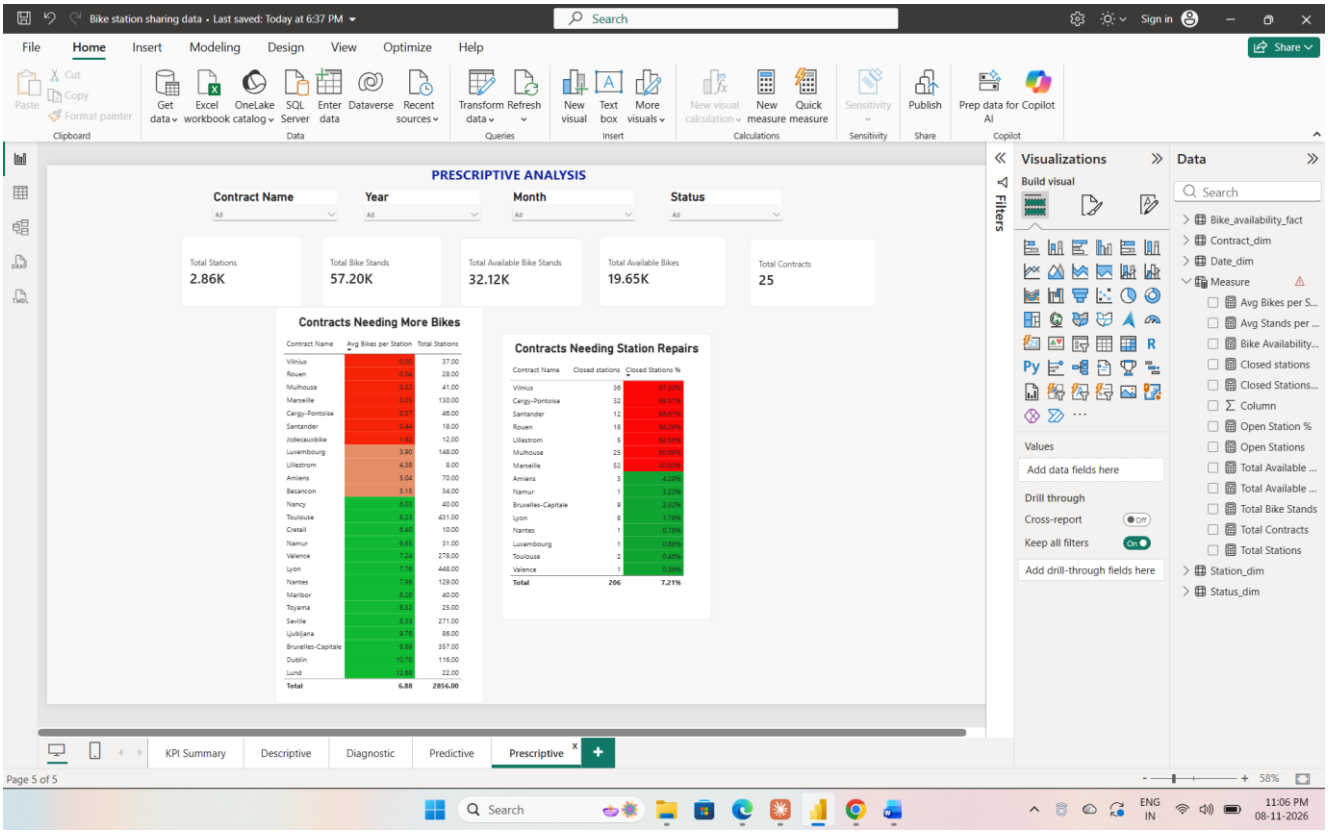
Descriptive Analysis



Diagnostic Analysis



Predictive Analysis



Prescriptive Analysis

9. Insights & Conclusions

Key findings

2.86K Total stations	**25** Contracts	**34.34%** Bike availability	**6.88** Avg bikes per station
2.65K Open stations	**206** Closed stations	**7.21%** Closed station%	**92.79%** Open Station %

- **Network size:** The top 10 contracts hold about 84% of all stations, led by Lyon (448) and Toulouse (431).
- **Availability:** Only 34.34% of bike stands hold a bike. A station has 6.88 bikes on average against 20.03 stands. Available bikes (19.65K) and empty stands (32.12K) add up to 51.77K of the 57.20K stands, so about 5.4K stands are not accounted for by either measure and should be checked against the source data.
- **Operations:** 92.79% of stations are open and 206 (7.21%) are closed.
- **Closures:** Marseille, Vilnius, Cergy-Pontoise, Mulhouse and Rouen account for 163 of the 206 closed stations (79%), and closures are not linked to network size.
- **Shortage:** Six contracts average under 0.5 bikes per station: Vilnius, Rouen, Mulhouse, Marseille, Cergy-Pontoise and Santander.
- **Trends:** Bikes per station show a slight but unstable upward trend, while closed stations show a slight downward trend with sharp spikes.

Descriptive analysis

Question 1: Which contracts have the most stations? *(Chart: Stations by Contract)*

- Lyon has the most stations (448), followed by Toulouse (431), Bruxelles-Capitale (357), Valence (278) and Seville (271).
- The numbers drop sharply after the top five, and Lillestrom has the fewest stations (8). The network is concentrated in a few large contracts.

Question 2: What share of the total stations do the top 10 contracts hold? *(Chart: Top 10 Contracts by Number of Stations)*

- The top 10 contracts hold 2,394 of the 2,856 stations, about 84% of the network.
- The tenth contract, Ljubljana (86), has only 19.2% of the stations of Lyon, the largest contract (448).
- The network size is therefore uneven across contracts, and a few large contracts hold most of the stations.

Question 3: Where are the bike stations located? *(Chart: Bike Station Locations)*

- The map shows that almost all stations are in Europe, mainly in western Europe, which matches the French and European contracts in the data.
- One small cluster appears in Japan (Toyama).
- So the network is mostly a European system.

Question 4: How many stations are open and how many are closed? *(Chart: Open vs Closed Stations)*

- Out of 2.86K stations, 2.65K (92.79%) are open and 0.21K (206 stations, 7.21%) are closed.
- The network is mostly operational, but the closed share is still large enough to need attention.

Diagnostic analysis

Question 5: Do contracts with more stations also have more closed stations? *(Chart: Stations vs Closed Stations by Contract)*

- No, Large contracts such as Lyon (448 stations, 8 closed) and Toulouse (431 stations, 2 closed) have very few closures.
- Smaller contracts such as Vilnius (36 closed), Cergy-Pontoise (32), Mulhouse (25) and Rouen (18) have many closures for their size.
- Marseille has the highest number of closed stations (52 out of 130).
- Closures therefore do not appear to depend on network size; they may reflect how each contract is run, which this data cannot confirm.

Question 6: Which contracts have the fewest bikes per station? *(Chart: Average Bikes per Station by Contract)*

- The overall average is 6.88 bikes per station.
- Lund (12.7), Dublin (10.7), Bruxelles-Capitale (9.9) and Ljubljana (9.8) are the highest.
- Vilnius (0.00), Rouen (0.04) and Mulhouse (0.32) are the lowest, followed by Marseille, Cergy-Pontoise and Santander at about 0.4 bikes per station.
- This shows that low availability is concentrated in a few contracts.

Question 7: Which contracts contribute most to the 206 closed stations? *(Chart: Closed Stations by Contract)*

- Marseille (52), Vilnius (36), Cergy-Pontoise (32), Mulhouse (25) and Rouen (18) together account for 163 of the 206 closed stations, about 79%.
- The remaining contracts add only small amounts each.
- So fixing these five contracts would solve most of the closure problem.

Predictive analysis

Question 8: Is the average number of bikes per station increasing or decreasing over time? *(Chart: Average Bikes per Station by Month)*

- The trend line rises slightly, from close to 0.5 bikes per station in early 2022 to about 3 by late 2025.
- The data is unstable, with peaks of about 11 in mid and late 2024 and a high of 7.9 in the latest month, while most other months stay below 3.
- The improvement is weak and not yet reliable.
- Each station has only one last-update date, so these trends show when stations last reported rather than true changes over time.

Question 9: What is the expected trend for closed stations? *(Chart: Closed Stations Trend by Month)*

- The trend line slopes slightly downward, from about 20 closed stations in early 2023 to about 12 by late 2025, so closures may fall slowly.
- Sharp spikes (52 in early 2023, 33 in mid-2025 and 48 in the latest month) show that closures are still unpredictable.
- Power BI's forecast could not run because some months have no data, so trend lines were used.

Prescriptive analysis

Question 10: Which contracts need more bikes first? *(Chart: Contracts Needing More Bikes)*

- Vilnius (0.00), Rouen (0.04), Mulhouse (0.32), Marseille (0.35), Cergy-Pontoise (0.37), Santander (0.44) and Jcdecauxbike (1.92) fall in the red group, with fewer than 2 bikes per station. Marseille is the top priority because it has 130 stations.
- Extra bikes only help where stations are open, so contracts with many closed stations (Question 11) need repairs first.
- Because each contract is a separate city system, bikes should be redistributed within each contract, using high-availability contracts such as Lund, Dublin, Bruxelles-Capitale and Ljubljana as the benchmark.

Question 11: Which contracts need station repairs first? *(Chart: Contracts Needing Station Repairs)*

- Vilnius (97.30% closed), Cergy-Pontoise (69.57%), Santander (66.67%), Rouen (64.29%), Lillestrom (62.50%) and Mulhouse (60.98%) have the highest share of closed stations and should be repaired first.
- Marseille has the most closed stations (52, or 40%), so it needs repairs in volume.
- The network average is 7.21%, so these contracts are far above the norm.

Bike Sharing Network Improvement Strategies

1. **Bike Redistribution:** Redistribute bikes within each contract, using the high-availability contracts (Lund, Dublin, Bruxelles-Capitale and Ljubljana) as the benchmark. Give priority to low-availability contracts such as Vilnius, Rouen, Mulhouse, Marseille, Cergy-Pontoise and Santander once their closed stations have been repaired and reopened.
2. **Station Maintenance:** Prioritise the repair and reopening of closed stations, starting with Vilnius, Cergy-Pontoise, Santander, Rouen, Lillestrom and Mulhouse (highest share of closed stations), followed by Marseille (highest number of closed stations).
3. **Performance Monitoring:** Monitor the closed-station rate monthly and establish an alert when it exceeds 10% to identify operational issues early.
4. **Future Data Analysis:** Incorporate revenue, distance and operating cost data to enable further analysis of revenue per kilometre and operational cost efficiency.

10. Conclusions

- This project demonstrates how Microsoft Power Query and Power BI can be used to transform raw bike-sharing data into meaningful insights.
- Through data cleaning, transformation, star-schema modelling and DAX calculations, a structured data model was developed to analyse station performance and bike availability.
- Interactive dashboards enabled descriptive, diagnostic, predictive and prescriptive analysis across different contracts.
- The analysis reveals that 92.79% of stations are operational, while overall bike availability is only 34.34%, with low availability concentrated in a few contracts, which points to an uneven distribution of bikes.
- The major operational challenges are concentrated in contracts such as Marseille, Vilnius, Cergy-Pontoise, Mulhouse and Rouen.
- The findings highlight the need for better bike redistribution, timely station maintenance and continuous performance monitoring to improve bike availability and overall network efficiency.